CAPACITIVE POWER TRANSFER | TECHNICAL SCHEMATIC

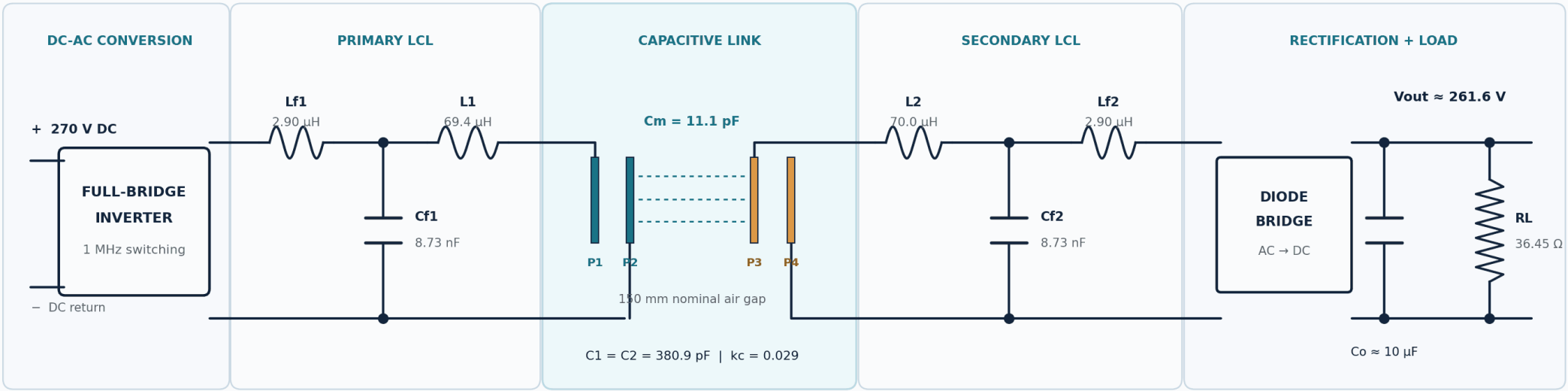
Double-sided LCL compensation • MATLAB numerical reproduction • Four-plate coupler

CPT-ENG-001

REV A | 26 OCT 2026

01 / POWER STAGE AND EQUIVALENT COUPLER INTERFACE

SYSTEM-LEVEL ELECTRICAL SCHEMATIC



READING NOTE Power transfer occurs through an electric field; no conductive connection crosses the plate interface.

02 / SIMULATION PARAMETERS AND VALIDATION

EXCITATION

Input supply	**270 V DC**
Switching frequency	**1 MHz**
Load	**36.45 Ω**

LCL NETWORK

Series filter inductors	**Lf1 = Lf2 = 2.90 µH**
Resonant inductors	**L1 = 69.4 / L2 = 70.0 µH**
Filter capacitors	**Cf1 = Cf2 = 8.73 nF**

FOUR-PLATE COUPLER

Equivalent capacitances	**C1 = C2 = 380.9 pF**
Mutual capacitance	**Cm = 11.1 pF**
Separation / coupling	**150 mm / 2.9%**

MATLAB MODEL

Numerical solver	**9 states; fixed-step RK4**
Time step / duration	**1 ns / 3 ms**
Simulated output	**261.6 V DC**

SCOPE / ENGINEERING DECLARATION

An original presentation drawing of the electrical topology independently reproduced and simulated in MATLAB.

Reference: H. Zhang et al., IEEE Transactions on Power Electronics 31(12), 2016, DOI 10.1109/TPEL.2016.2520963.

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